

16.1 RFID			
EPC Code write(UHF)			ESC+IP0
Command	ESC	IP0	Parameter
HEX code	<1B> ₁₆	<49> ₁₆ <50> ₁₆ <30> ₁₆	See under "Parameters".
Default Value	Nil		
Valid range and term of command	When turning off the power switch		The set command is not maintained.
	Valid range within item		The set parameter becomes invalid.
	Valid range between items		The set parameter becomes invalid.

[Function]

This command writes the EPC code and other data to the EPC corresponding RFID tag. The standard format (three-character parameter identifier) is used when using the memory area extended for each IC chip manufacturer. Available in M6e module installed printers.

[Format]

[Standard format]

<IP0>e:h(pcw:xxx)(epc:xxxx...xxx)(usr:xxx...xxx)(,usa:dddd,aaaaa...aaaa)(,lck:bbbb);

[Conventional format]

<IP0>e:a,c,d,s:d,dddddddddd(o:xxx)(,m:bbbb)(,k:xxxxxxx)(,a:xxxxxxx)(,p:xxxxxxx);
[SSCC96 code conversion]
 <IP0>e:c,f,d,c:d,t,d,dddddddddd(n:dddddddddd(o:xxx)(,m:bbbb)(,k:xxxxxxx)(,a:xxxxxxx)
 (,p:xxxxxxx);
[SGTIN96 encoding]
 <IP0>e:z,(d:xxxxxxxxxxxxxxxxxxxxxxx)(,u:xxxxxxxxxxxxxxxxxxx)(,k:xxxxxxx)(,a:xxxxxxx)
 (,p:xxxxxxx)(,o:xxx)(,m:bbbb)(,k:xxxxxxx)(,a:xxxxxxx)(,p:xxxxxxx);
[Free mapping]

Note: Typical parameters are shown in the format examples above. Refer to the next item for details on the parameters.

[Conventional format] is solely for maintaining compatibility with models before CL4NX. If you are going to use a new command from CL4NX and beyond, please create with [Standard format].

●Parameter

EPC encode type parameter			
e:	EPC encode Type Specify the parameter right after <IP0>. The parameter cannot be omitted.	h	[Standard format] Use the three-character parameter identifier after the declaration. The memory area extended for each IC chip manufacturer can be extended for other than the standard Gen 2 tag.
		a	[Conventional format] SSCC96 code conversion Use the "one-character parameter identifier". When the Company prefix/Serial reference/Filter value is received, and then converted to an SSCC96 compliant code, and written to the tag.
		c	[Conventional format] SGTIN96 code conversion Use the "one-character parameter identifier". When the Company prefix/Serial reference/Filter value is received, and then converted to an GTIN96 compliant code, and written to the tag.
		z	[Conventional format] Free mapping Use the "one-character parameter identifier". When data is written as is without performing code conversion.

Standard format

This is the extended format for the tag from IC chip manufacturer having over 96 bits EPC area.

<IP0>e: Write these parameters after specifying h.

<IP0>e: The order of the parameters after specifying h doesn't matter.

Three-character parameter identifier

Gen 2 basic operation parameters (can be omitted)		
pcw:	PC rewrite	The PC area contains length information for controlling the number of digits of data to be recorded to the EPC. You can change the number of digits of the EPC data by specifying multiples of four from 4 to 124 in decimal code to this parameter. Specifying this parameter is not necessary in most cases because the length information is written to the PC automatically by judging from the data length specified by epc.; if this parameter is not specified. (Maximum value of the length information of PC is 124 digits.)
afi:	AFI write data	Specify the number from 000 to 1FF in HEX code. Synthesize data together with current PC length information and write.
epc:	EPC write data	Specify 4 to 124 digit number in HEX code corresponding to the traditionally written data in the EPC area. Digits can be varied in multiples of four. When the character string is longer than the digits specified by pcw., a parameter error occurs, and when the character string is shorter, the blanks are filled with zeros. You may also use the space, tab, line feed in data separator, but these codes will be skipped.
usr:	USER memory write	Specify numbers to write to USER area in hexadecimals. It must be specified in digits that are multiples of 4. The numbers should be within 4-128 digits, but the maximum number of digits depends on the tag in use. You may also use the space, tab, line feed in data separator, but these codes will be skipped.
usa:	USER memory ASCII write	Specify numbers to write to USER area in ASCII. Data that includes Control code (00H-1FH) can be written as well. Note that size information of the data area separated with comma is required in front of the data. Example: If the data size is 64, specify usa:64,abcde... The digits should be a multiple of 2. The numbers should be specified within 2- 64 digits, but the maximum number of digits depends on the tag in use.

Operation parameters in standard format (can be omitted)		
lck:	Lock	Specify the area to be locked by five bits in binary code such as lck:10101. The display order is USER, TID, ACCESS, KILL, EPC, from the top. If the access password is specified, it results in a password lock, and if the access password is not specified, it results in a permanent lock.
apw:	Access password	To perform locking or read protecting with a password, specify an eight-character HEX code.
psw:	Unlocking password	To access an area on which a password lock or password read lock is applied, specify the same number sequence as that used when locking. Specify unlocking area with lck:.
kpw:	Kill code	Specify the kill code with eight-character HEX code. You cannot execute the kill by writing the code only.

NXP G2XL dedicated secure parameters (can be omitted) (Cannot be used in a Gen2 chip other than NXP G2XL.)		
rlk: G2X,	Read protection	Specify 0 or 1 to indicate whether read protection is to be set or canceled. 1 for set and 0 for reset. When read protection is set, the reading of all areas becomes restricted. You must concurrently use an access password other than "00000000", and for reset, specify the same number sequence as the access password used during setting, in the cancellation password. (dedicated for NXP G2XL)
eas: G2X,	EAS bit data	Specify 0 or 1 as the value of the EAS bit. 1 for set and 0 for reset. You must concurrently use an access password other than "00000000", and for reset, specify the same number sequence as the access password used during setting, in the cancellation password. (dedicated for NXP G2XL)

NXP G2iM/G2iL dedicated secure parameters (can be omitted) (Cannot be used in a Gen2 chip other than NXP G2iM/G2iL.)		
rlk: G2I,	Read protection	Specify 0 or 1 to indicate whether read protection is to be set or canceled. 1 for set and 0 for reset. When read protection is set, the reading of all areas becomes restricted. You must concurrently use an access password other than "00000000", and for reset, specify the same number sequence as the access password used during setting, in the cancellation password. (dedicated for NXP G2iM/G2iL)
eas: G2I,	EAS bit data	Specify 0 or 1 as the value of the EAS bit. 1 for set and 0 for reset. You must concurrently use an access password other than "00000000", and for reset, specify the same number sequence as the access password used during setting, in the cancellation password. (dedicated for NXP G2iM/G2iL)

Feed control parameters (can be omitted)		
fsw:	Feed without printing	Usually, for RFID write item without print data, only writing of RFID data is performed, and the label is not fed. (Except for cases where the tag offset is set to other than default.) When "1" is specified in this parameter, the label is fed even if the item does not contain print data. When "0" is specified for this parameter, the label is not fed for the item without print data, and the operation is the same as when fsw: is not specified. If don't feed, Short buzzer will sound 2 times when write is successful. Short buzzer will the sound 6 times when it fails to write. In addition, if 2 is specified for the item including print data, a write test can be performed without consuming a label as only write is executed without feed/print. Valid range : 0-2

SSCC96/SGTIN96 standard code conversion parameters		
enc:	Declaration of code standard conversion	Declare when executing conversion according to the code standard. SSCC96 and SGTIN96 can be specified. If you specify 'a', it indicates SSCC96, and if you specify 'c', it indicates SGTIN96. The converted data is written in the EPC area. Therefore, you can no longer use the epc: parameter identifier Valid range: a, c
flt:	Filter value	Specify the filter value with the number from 0 to 7. Refer to the filter value standard table described later.
com:	Company prefix	Specify the company prefix with a 6 to 12-digit decimal number.
srl:	Serial reference	In the case of SSCC96, specify the serial reference with a 5 to 11-digit decimal number together with the number sequence specified in com:, so that the value is 17 digits or less. Even if the value does not reach 17 digits, leave it as is since the upper digits are filled with a zero inside the printer. In the case of SGTIN96, specify the serial number with 12-digit fixed decimal number. Specify 274877906943 or a smaller number. It becomes a parameter error in the case of less than 12 digits.
itm:	Item reference (Only for SGTIN96)	Specify the item reference with a 1 to 7-digit decimal number together with the number sequence specified in com:, so that the value is 13 digits or less. Even if the value does not reach 13 digits, leave it as is since the upper digits are filled with a zero inside the printer.

Coding examples of the standard format (three-character parameter identifier)

[Coding example 1]

When data is written in an arbitrary length in the EPC region in the IC chip manufacturer extension specifications tag

<A>

<V>50<H>50<XM>SECURE RFID EPC DATA WRITE

<IP0>e:h,epc:01234567890123456789012345678901; (in digits that are multiples of 4)

<Q>1

<Z>

[Coding example 2]

When EPC and user memory are written in a batch in the IC chip manufacturer extension specifications tag, and permanent locking is performed

<A>

<V>50<H>50<XM>SECURE RFID EPC & USER DATA WRITE

<IP0>e:h,epc:01234567890123456789012345678901,usr:012345678.....5676789012345678901,lck:10001;

<Q>1

<Z>

[Coding example 3]

When data is written in EPC by specifying the SSCC96 standard code conversion in the Gen2 tag, and permanent locking is performed

<A>

<V>50<H>50<XM>SECURE RFID SSCC96 DATA WRITE

<IP0>e:h,enc:a,flt:0,com:12345678,srl:012345678,lck:00001;

<Q>1

<Z>

[Coding example 4]

When data is written in EPC by specifying the SGTIN96 standard code conversion in the Gen2 tag

<A>

<V>50<H>50<XM>SECURE RFID SGTIN96 DATA WRITE

<IP0>e:h,enc:c,flt:0,com:123456,itm:0123456,srl:012345678901;

<Q>1

<Z>

Conventional format

The format for Gen2(Class1Generation2) tag only.

Use the standard format for the tag from the IC chip manufacturer having an EPC area of more than 96 bits.

<IP0>e: Write these parameters after specifying “ * ”.

Conventional format: One-character parameter identifier (SSCC96 code conversion / SGTIN96 code conversion / free mapping)

EPC data parameter			
d:	EPC data	Specify the EPC data. The restrictions on the number of digits based on each EPC encoding type are as follows:	
		When SSCC96 code conversion is specified <IP0>e:a,..	A 17-digit decimal number sequence in which the company prefix and serial reference are joined together
		When SGTIN96 code conversion is specified <IP0>e:c,..	A 13-digit decimal number sequence in which the company prefix and item reference are joined together
		When free mapping is specified <IP0>e:z,..	A hexadecimal string containing 24 or less number of digits in multiples of four, which is written without conversion in the EPC area. Use the standard format in an application exceeding 24 digits (96 bits). You may also use the space, tab, line feed in data separator, but these codes will be skipped.

SSCC96/SGTIN96 conversion parameter		
f:	filter value	Refer to the filter value and the filter value standard table described later.
c:	company prefix	Number of digits of the company prefix A number from 6 to 12. Number of digits of the company prefix in the first half of the data specified in the d parameter.
s:	serial reference	Number of digits of Serial reference (only when selecting SSCC96) Number of digits of the serial reference in the latter half of the data specified in the d parameter. Specify a number from 5 to 11, so that when combined with c, the resultant value is 17.
t:	item reference	Number of digits of Item reference (only when selecting SGTIN96) Specify a number from 1 to 7, so that when combined with c, the resultant value is 13.
n:	serial number	Serial number 12 digits (fixed) (only when selecting SGTIN96) Specify 274877906943 or a smaller number.

Feed control parameters (can be omitted)		
g:	Feed without printing	Usually, for RFID write item without print data, only writing of RFID data is performed, and the label is not fed. (Except for cases where the tag offset is set to other than default) When "1" is specified in this parameter, the label is fed even if the item does not contain print data. When "0" is specified for this parameter, the label is not fed for the item without print data, and the operation is the same as when g: is not specified.

Gen 2 memory bank operation parameters (can be omitted)														
u:	User Memory HEX	Specify numeric string to write to USER area in hexadecimals. Specified digits must be multiples of 4. This can be specified in the range of 4 to 128 digits, but the maximum number of digits depends on the used tag. You may also use the space, tab, line feed in data separator, but these codes will be skipped.												
v:	User Memory Ascii	Specify numeric string to write to USER area in ASCII data. It is necessary to have data area and data size information separated with a comma before the data. For example in specifying a data size as 16, specify as follows: V:16,abcd... Specified digits must be multiples of 2. This can be specified in the range of 2 to 64 digits, but the maximum number of digits depends on the used tag.												
o:	AFI code	Application Family Identifier. Specify a hexadecimal number from 0H to 1FFH. To write as AFI, specify the ninth bit (MSB) as 1.												
m:	Locking	5-digit binary number Set the bit of the bank to be locked as 1. If you specify the access code, it results in a "password lock", and if you omit the access code, it results in a "permanent lock". <table><tr><td>Bit</td><td>Section to be locked</td></tr><tr><td>bit0 (LSB)</td><td>EPCdata</td></tr><tr><td>bit1</td><td>KillCode</td></tr><tr><td>bit2</td><td>AccessCode</td></tr><tr><td>bit3</td><td>TIDdata</td></tr><tr><td>bit4 (MSB)</td><td>UserMemory</td></tr></table> Example: If only the EPCdata area is locked "m: 00001"	Bit	Section to be locked	bit0 (LSB)	EPCdata	bit1	KillCode	bit2	AccessCode	bit3	TIDdata	bit4 (MSB)	UserMemory
Bit	Section to be locked													
bit0 (LSB)	EPCdata													
bit1	KillCode													
bit2	AccessCode													
bit3	TIDdata													
bit4 (MSB)	UserMemory													
k:	kill code	8-digit HEX code Used when writing the kill code. Specify other than "00000000" in an eight-digit HEX number sequence configured by numbers from 0 to 9, and alphabets from A to F. The purpose of this parameter is only to write the kill code. You cannot kill the tag actually.												
a:	access code	This is an eight-digit number sequence specified when applying the password lock. You can use an eight-digit HEX number sequence configured by numbers from 0 to 9, and alphabets from A to F. To apply the password lock, specify other than "00000000". If you omit this parameter, and specify m: Locking, a permanent lock is applied on the specified area.												
p:	password code	This is an eight-digit number sequence used for rewriting the tags on which a password lock has already been applied. It is an eight-digit HEX number sequence configured by numbers from 0 to 9, and alphabets from A to F that is used when writing in a tag on which a password lock has been applied. Specify the same number sequence as the access code used at the time of locking. By specifying a number sequence same as the access code specified at the time of password lock in the "password code", and specifying "00000000" in the "access code", the tag on which a password lock has been applied can be returned to an open state.												

Coding examples of one-character parameter identifiers
(SSCC96 code conversion / SGTIN96 code conversion / free mapping)

[Coding example 1] Country code 49 /Company code 12345 / Serial no. 789

When SSCC 96 data is written to a Gen2 corresponding tag

<A>

<V>50<H>50<XM>49123450000000789

<IP0>e:a,c:7,s:10,f:1,d:49123450000000789;

<Q>1

<Z>

[Coding example 2] Country code 49 /Company code 12345 / Item 67890 / Serial no. 789

When SGTIN96 data is written to a Gen2 corresponding tag

<A>

<V>50<H>50<XM>491234506789000000000789

<IP0>e:c,c:7,t:6,f:1,d:4912345067890,n:000000000789;

<Q>1

<Z>

[Coding example 3]

When free mapping data is written in the EPC area of a Gen2 corresponding tag

<A>

<V>50<H>50<BD>101080*301803CB4F48B38000000001*

<IP0>e:z,d:301803CB4F48B38000000001;

<Q>1

<Z>

[Coding example 4]

When data is written in free mapping in the EPC area of a Gen2 corresponding tag, data is written in the user area, and at the same time, a permanent lock is applied to both the EPC and USER areas

<A>

<V>50<H>50<BD>101080*0123456789ABCDEF01234567*

<IP0>e:z,d:0123456789ABCDEF01234567,u:0123456789ABCDEF0123456789ABCDEF0123456789ABCDEF,m:10001;

<Q>1

<Z>

[Coding example 5]

When data is written in free mapping in the EPC area of a Gen2 corresponding tag, and a password lock is applied to the EPC area. When the password is set as 12345678

<A>

<V>50<H>50<BD>101080*0123456789ABCDEF01234567*

<IP0>e:z,d:0123456789ABCDEF01234567,a:12345678,m:00101;

<Q>1

<Z>

[Supplemental explanation]

1. This command can be used only once between the data transmit start <A> to the data transmit end <Z>.
2. When print data is to be included beside EPC code write, multiple labels can be specified. If there is no print data as in Example 2, media is not fed. If a blank media is to be fed after EPC code write, specify feeding without print as in Example 3.

[Example 1] If print data is included, the QTY can be specified.

```
<A>
<V>50<H>50<XM>0123456789ABCDEF01234567
<IP0>e;z.d:0123456789ABCDEF01234567;
<Q>1
<Z>
```

[Example 2] If you specify EPC code write without print data, the media is not fed, and QTY cannot be specified.

```
<A>
<IP0>e;z.d:0123456789ABCDEF01234567;
<Z>
```

[Example 3] If there is no print data, but you have specified feeding without print in EPC code write, the media is fed, and QTY can be specified.

```
<A><IP0>e;z.d:0123456789ABCDEF01234567.q:1; <Q>2<Z>
or
<A><IP0>e:h,epc:0123456789ABCDEF01234567,fsw:1; <Q>2<Z> (for the standard format)
```

	Example 1	Example 2	Example 3
When EPC code write is successful	Print, feed	Don't feed Two short beeps	Feed a blank label
When EPC code write has failed	Print tag error Retry operation	Six short beeps	Print tag error Retry operation

* If you have set TAG OFFSET to other than the default value, or if you have specified temporary tag offset, feeding is performed in Example 2 as well.

3. The buzzer sounds as described below.

If an EPC code write that does not accompany print data is specified, the buzzer sounds to notify you whether the EPC code write succeeded.

EPC code write successful: Two short beeps

EPC code write unsuccessful: Six short beeps

Note that the buzzer sounds for EPC code write success or failure vary depending on the "Sound" > "Error Sound" settings in the System menu.

4. Unused parameters can be omitted.

5. For details on the filter value to be used with SSCC96/SGTIN96, refer to the table below according to the purpose of use.

Filter value standard table

Filter value	SSCC96	SGTIN96
0	All Others	All Others
1	Reserved	Point of Sale (POS) Trade Item
2	Full Case for Transport	Full Case for Transport
3	Reserved	Reserved
4	Reserved	Inner Pack Trade Item Grouping for Handling
5	Reserved	Reserved
6	Unit Load	Unit Load
7	Reserved	Unit inside Trade Item or component inside a product not intended for individual sale

6. SSCC96 code conversion

This function is used to generate a code according to the SSCC96 standard when a company prefix, serial reference, and filter value are applied. For the time being, the usage method is explained by using an example in which the following numbers have been applied. All numbers are in decimals.

Company prefix: 4011661

Serial reference: 3000000001

Filter value: 0 (Select the filter value from the filter value standard table described earlier according to the purpose of use. In this example, the filter value is 0 from "All Others")

According to the standard, the company prefix and serial reference are combined together to result in 17 digits. Of these 17 digits, the proportion of the company prefix and the serial reference can be selected from the seven combinations described in the table below.

Partition value	Number of digit of the company prefix	Number of digits of the serial reference
0	12	5
1	11	6
2	10	7
3	9	8
4	8	9
5	7	10
6	6	11

In the case shown in this example, the following combination is assumed:

Number of digits of the company prefix: 7

Number of digits of serial reference: 10

The partition value is 5, but since this value is generated within the PC, it need not be specified.

Under this condition, if the format of EPC code write <IP0> is applied, it results in the following:

When the standard format is used

<IP0>e:h,enc:a,com:4011661,srl:3000000001,flt:0;

Description of each parameter

e:h,----- Specifies the standard format

enc:a,----- Specifies SSCC96 code conversion

com:4011661,----- The company prefix is **4011661**

srl:3000000001,-- The serial reference is **3000000001** 17 digits when combined with com:

flt:0,----- The filter value is All Others

;------ End terminal

When the one-character parameter identifier in the conventional format is used

<IP0>e:a,c:7,s:10,f:0,d:40116613000000001;

Description of each parameter

e:a,----- Specifies SSCC96 code conversion

c:7,----- The company prefix is 7 digits

s:10,----- The serial reference is 10 digits 17 digits when combined with c:

f:0,----- The filter value is All Others

d:40116613000000001 –Number in which the company prefix and serial reference are joined together It is 17 digits.

;------ End terminal

When these commands are executed, the above parameters are converted to SSCC96 code 3114F4DA34B2D05E01000000 and 24 characters (18 characters in actuality), and are written in a tag.

7.SGTIN96 code conversion

This function is used to generate a code according to the SGTIN96 standard when a company prefix, item reference, filter value, and serial number are applied. For the time being, the usage method is explained by using an example in which the following numbers have been applied. All numbers below are in decimals.

Company prefix: 003885

Item reference: 4006606

Filter value: 0 (Select the filter value from the filter value standard table described earlier according to the purpose of use. In this example, the filter value is 0 from "All Others")

Serial number: 000000000001 (fixed as 12 digits)

According to the standard, the company prefix and item reference are combined together to result in 13 digits. Of these 13 digits, the proportion of the company prefix and the item reference can be selected from the seven combinations described in the table below.

Partition value	Number of digit of the company prefix	Number of digit of the item reference
0	12	1
1	11	2
2	10	3
3	9	4
4	8	5
5	7	6
6	6	7

In the case shown in this example, the following combination is assumed:

Number of digits of the company prefix: 6

Number of digits of item reference: 7 If both items combined together do not form 13 digits, 0 is entered before the number to fill up the lacking number of digits.

The partition value is 6, but since this value is generated within the PC, it need not be specified.

Under this condition, if the format of <IP0> EPC code write is applied, it results in the following:

When the standard format is used

<IP0>e:h,enc:c,com:003885,itm:4006606,flt:0,srl:000000000001;

Description of each parameter

e:h,----- Specifies the standard format

enc:c,----- Specifies SGTIN96 code conversion

com:003885,----- The company prefix is 6 digits

itm:4006606,----- The item reference is 7 digits 13 digits when combined with com:

flt:0 ----- The filter value is All Others

srl:000000000001--- Serial number, fixed at 12 digits 274877906943 or a smaller number.

;------ End terminal

When the one-character parameter identifier in the conventional format is used

<IP0>e:c,c:6,t:7,f:0,d:0038854006606,n:000000000001;

Description of each parameter

e:c,----- Specifies SGTIN96 code conversion

c:6,----- The company prefix is 6 digits

t:7,----- The item reference is 7 digits 13 digits when combined with c:

f:0,----- The filter value is All Others

d:0038854006606, -Number in which the company prefix and item reference are joined together It is 13 digits.

n:000000000001--- Serial number, fixed at 12 digits 274877906943 or a smaller number.

;------ End terminal

When these commands are executed, the above parameters are converted to SGTIN96 code 301803CB4F48B38000000001 and 24 characters, and are written in a tag.